APPARATUS FOR A RECONFIGURABLE-WIDTH OPTICAL BUS

Reinaldo Javier Aguiar Marcano

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Title: APPARATUS FOR A RECONFIGURABLE-WIDTH OPTICAL BUS

Inventor: Reinaldo Javier Aguiar Marcano, a citizen of the United States, residing at 2302 Britton Ridge Drive, Katy, TX 77494.

Cross-Reference to Related Applications

This application claims the benefit of and priority to U.S. Patent Application No. 19/559,705, the contents of which are hereby incorporated by reference in their entirety.

Abstract

An apparatus for an optical data bus that can dynamically reconfigure its effective bit-width based on computational demands. The apparatus comprises a plurality of parallel optical channels and a bus controller. The controller is configured to logically group and ungroup the optical channels into one or more logical buses of varying widths. For a massively parallel task, the controller can group all available channels into a single, ultra-wide bus (e.g., 1024-bit). For a task requiring multiple, smaller, independent data streams, the controller can partition the same physical channels into several narrower, concurrent buses (e.g., four 256-bit buses). This allows for workload-specific optimization of the physical data path in real-time.

Background of the Invention

Traditional electrical computer buses have a fixed physical width (e.g., 64-bit). This width is a static design choice that cannot be altered once the hardware is manufactured. This represents a compromise, as different computational tasks have different data path requirements. A task that processes a large volume of small data packets may be better served by multiple independent, narrower buses, while a high-performance computing task may need the widest possible bus. A fixed-width bus is therefore suboptimal for a system designed for general-purpose or varied workloads.

Summary of the Invention

The present invention is an apparatus for a reconfigurable-width optical bus that provides a dynamic and software-defined physical data path. The apparatus leverages a large array of parallel optical channels (pixels) and a programmable bus controller to create logical buses of variable width.

The bus controller, responding to commands from the operating system or application software, can partition the physical optical channels into various configurations. For example, a 44-pixel optical interface could be configured by the controller in multiple modes. In "Maximum Bandwidth Mode," all 44 pixels are grouped to form a single logical bus with a width over 1024 bits. In "Multitasking Mode," the controller could partition the same 44 pixels into four independent 11-pixel buses, each providing a 256-bit data path. This allows the bus architecture to be tailored in real-time to the specific needs of the current workload, maximizing efficiency and performance.

Statement of License Grant to the United States Government

The inventor, Reinaldo Javier Aguiar Marcano, hereby grants to the United States Federal Government and its military and defense agencies a perpetual, royalty-free, non-exclusive license to practice and use the invention described herein for any and all governmental, defense, and non-commercial purposes. This license explicitly excludes any and all state and local governments.

Brief Description of the Drawings

FIG. 1 is a conceptual diagram showing the bus in two different configurations: a single, ultra-wide mode and a partitioned, multi-bus mode.

Detailed Description of the Invention

Referring to **FIG. 1**, the reconfigurable nature of the apparatus is shown. A physical interface comprising a plurality of optical channels (101) is managed by a bus controller (102). In a first configuration, "Maximum Bandwidth Mode" (110), the controller receives a command (111) and logically groups all physical channels together to form a single, ultra-wide logical bus (112), for example, 1024 bits wide. In a second configuration, "Multitasking Mode" (120), the controller receives a different command (121) and partitions the same physical channels into multiple, independent, narrower logical buses (122a, 122b, 122c, 122d), for example, four concurrent 256-bit buses.

Claims

What is claimed is:

1. A reconfigurable data bus apparatus, comprising:
    - a plurality of parallel optical channels; and
    - a bus controller communicatively coupled to the plurality of optical channels;
    - wherein the bus controller is configured to dynamically partition the plurality of optical channels into a variable number of logical buses, each logical bus having a variable bit-width.
2. The apparatus of claim 1, wherein the controller is configured to, in a first mode, group all optical channels to form a single logical bus of a first bit-width.
3. The apparatus of claim 2, wherein the controller is configured to, in a second mode, partition the optical channels to form at least two independent logical buses, each having a second bit-width that is less than the first bit-width.
4. The apparatus of claim 1, wherein the controller reconfigures the bus in response to a command from an operating system or an application.

5. The apparatus of claim 1, wherein the parallel optical channels are pixels in a two-dimensional array.

6. The apparatus of claim 1, wherein the bus is an implementation of the optical interface described in U.S. Patent Application No. 19/559,705.

Drawings

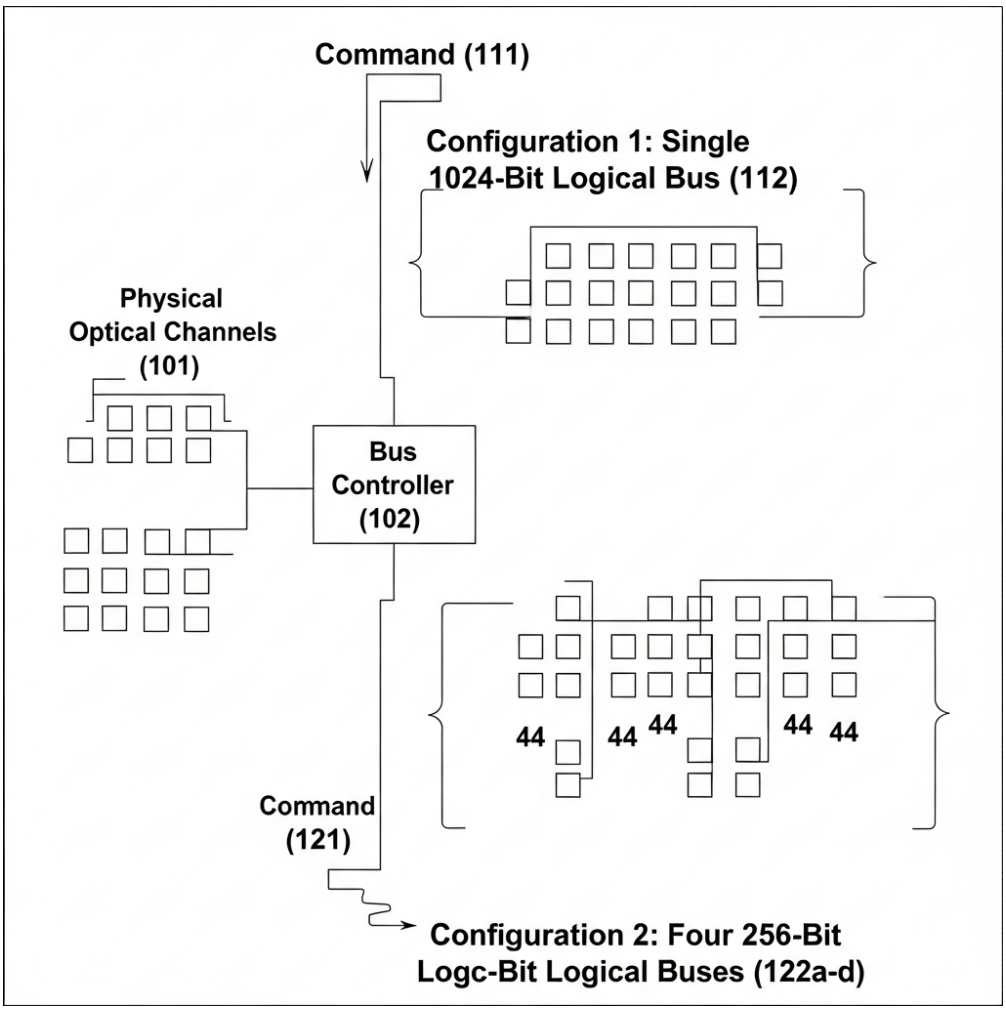


Figure 1: Conceptual diagram showing the bus in two different configurations: a single, ultra-wide mode and a partitioned, multi-bus mode.